ES 117: World of Engineering (2026)

Individual Ideation Assignment: Problem to Product

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Submission format: PDF only (via Google Classroom)
Deadline: January 30, 2026 (23:59 IST)
Submission mode: Individual

1. Assignment Objective

The purpose of this assignment is to evaluate how you think as an engineer when faced with an open-ended problem. The emphasis is not on arriving at a fully developed product or business plan, but on demonstrating clarity of thought, structured ideation, and the ability to reason from a problem toward a plausible solution direction.

Through this exercise, you are expected to identify a real-world problem, reflect on why it is worth solving, and communicate how engineering thinking can be applied to address it. The assignment values the quality of reasoning, motivation, and coherence over polish or completeness.

2. General Instructions

1. You are required to identify and frame your own problem. No predefined problem statements will be provided. The problem may arise from personal experience, observation of society or infrastructure, environmental concerns, digital systems, or any other context where engineering intervention is meaningful.

2. The ideation structure provided in this document is intended as guidance rather than a rigid template. You are not expected to fill every section. You may merge sections, omit those that are not relevant to your idea, or reorganize the flow entirely if doing so improves clarity. You are also encouraged to propose a better structure if you believe it communicates your thinking more effectively.

3. The most important aspect of your submission is the logical progression of ideas. The reader should be able to follow how you moved from identifying a problem to proposing a solution direction. Clear motivation, evidence of exploration, and internal consistency across slides are essential.

4. You are not expected to build a working prototype, produce detailed technical drawings, or present a complete business or financial plan. Approximate reasoning, high-level arguments, and qualitative assessments are sufficient, provided they are thoughtful and well-articulated.

3. Expected Deliverable

You are required to submit an ideation deck in PDF format through Google Classroom. The recommended length is between eight and fifteen slides. Submissions shorter than this are acceptable if the thinking is clear and sufficiently developed.

The deck should clearly convey the problem you are addressing, how you arrived at the idea, and the direction of your proposed solution. Visual clarity and concise explanations are encouraged.

4. Suggested Ideation Structure

The following sections outline a possible structure for your ideation deck. These sections are indicative and need not all be included.

4.1. Problem Statement

Clearly describe the problem you want to address. Indicate who is affected, in what context the problem arises, and why it is problematic. Avoid vague or overly broad statements.

4.2. Motivation

Explain why this problem matters and why you chose to work on it. This may include personal experiences, observations, or broader societal relevance. This section helps the reader understand your intent and perspective.

4.3. Origin of the Idea

Describe how you arrived at this problem and idea. This could involve everyday observations, recurring frustrations, exposure to data or news, or insights from discussions. The goal is to make your ideation process transparent.

4.4. Existing Gaps

Discuss why the problem has not been adequately solved so far. Consider technical limitations, cost constraints, usability issues, policy barriers, or lack of awareness. Avoid generic statements and aim for specific reasoning.

4.5. Proposed Solution Direction

Outline your core idea at a high level. This does not need to be a final product. Explain what kind of solution you are proposing and where engineering plays a central role.

4.6. Existing Solutions or Alternatives

Briefly describe existing products, systems, or approaches that attempt to address the problem. Explain where they fall short and what gap your idea attempts to fill.

4.7. Target Users

Identify the primary users or beneficiaries of your proposed solution. If relevant, distinguish between different user groups and note any constraints they may face, such as cost, accessibility, or technical literacy.

4.8. Impact and Scale

Explain what changes if your idea works. Describe the potential impact and whether it is local, regional, or scalable. Quantitative estimates are optional; clear reasoning is not.

4.9. Feasibility Snapshot

Provide a high-level assessment of feasibility. Mention key technical components, assumptions, and major challenges you anticipate. This section should demonstrate realism rather than certainty.

4.10. Optional Extensions

You may include additional sections such as a rough timeline, process flow, ethical considerations, sustainability aspects, or possible future extensions, if they add value to your narrative.

5. Evaluation Criteria

Submissions will be evaluated based on clarity of problem formulation, depth of ideation, relevance of engineering thinking, coherence of the overall narrative, quality of communication, and originality of thought.

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